# Yike Niu

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## EDUCATION

**Tsinghua University, Weiyang College** 2024 – 2028 (Expected)
*Mathematics and Physics + Mechanical Engineering* GPA: 3.9/4.0
**Selected coursework:** Calculus (4.0), Linear Algebra (4.0), Data Structures and Algorithms (4.0), Mechatronic Measurement and Intelligent Control (4.0).

## RESEARCH INTERESTS

- **Robot learning:** Efficient, reliable manipulation through visuomotor imitation learning.
- **Policy improvement:** Reinforcement learning, adaptation, and generalization through interaction.
- **Generalist robotics:** Vision-language-action models for complex, long-horizon manipulation.

## PUBLICATIONS AND PREPRINTS

Jiale Cao*, **Yike Niu***, Zhengrong Xue, and Huazhe Xu. **AdaTempo: Learning Shared Relative Tempo from Demonstrations for Faster Robot Manipulation.**
Preprint, 2026. Submitted to ICRA 2027. arXiv:2610.02706 * *Equal contribution.*

Mingju Gao, Kaisen Yang, …, **Yike Niu**, …, Hao Zhao. **PAM: A Pose–Appearance–Motion Engine for Sim-to-Real HOI Video Generation.**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. Paper

## RESEARCH EXPERIENCE

**AdaTempo | Robot Learning and Manipulation** Aug 2025 – Present
*Advisor: Prof. Huazhe Xu, Tsinghua University*

- **Method & Training**: Contributed to method design, demonstration processing, policy training, and hyperparameter tuning.
- **Simulation**: Led simulation experiments on robomimic and ALOHA, including ACT and Diffusion Policy comparisons and systematic ablation studies.
- **Real-World Evaluation**: Contributed to real-robot system setup, task design, data collection, and experimental evaluation across diverse manipulation tasks.
- **Outcome**: Co-authored an ICRA 2027 submission evaluating AdaTempo across 11 manipulation tasks, demonstrating up to 3.57× execution speedup over demonstrations.

**PAM | Sim-to-Real Hand–Object Interaction Video Generation** Sep – Dec 2025

- Processed and converted experimental data for sim-to-real hand–object interaction video generation.
- Contributed to qualitative analysis, result selection, and figure preparation for the CVPR 2026 paper.

## SKILLS

**Robot learning:** ACT, Diffusion Policy, robomimic, ALOHA simulation tasks, policy training and evaluation.
**Programming and ML:** Python, C++, PyTorch; MATLAB (basic).
**Platforms and tools:** ALOHA, Franka, Linux, SolidWorks (CAD), LaTeX.

## HONORS & AWARDS

**Comprehensive Excellence Scholarship** 2025 academic year
**Academic Excellence Scholarship** 2025 academic year
**Third Prize —** Weiyang City AI Agent Competition (Team Leader) May 2026
**Second Prize —** Chinese Mathematics Competitions Dec 2025
**Third Prize —** Regional University Physics Competition, Physics Division Dec 2025
**Second Prize —** Beijing College Student Mathematics Competition Nov 2025

## LANGUAGES & INTERESTS

**Languages:** Chinese; English (TOEFL iBT 110/120).
**Interests:** Chinese cooking, baking, fitness, and basketball.